

Development of Exercise Intervention Program for Gait and Posture Improvement in Extended-Reality Environment

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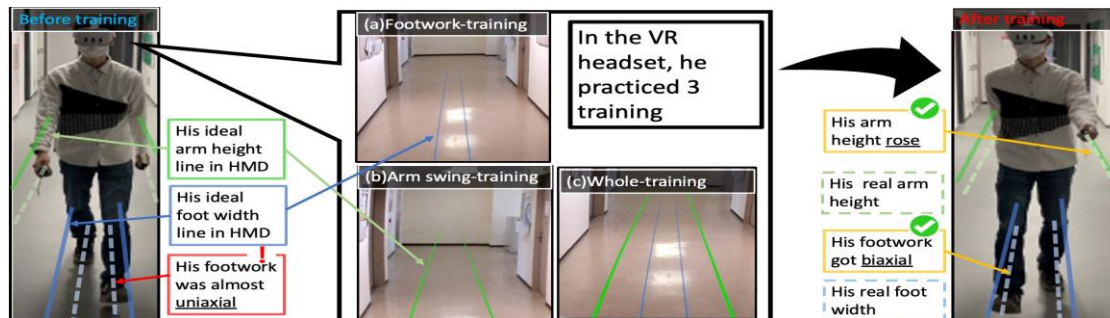


Figure 1: Image of the gait improvement contents in the XR space developed in this study and an example of gait improvement

ABSTRACT

In this study, we developed an exercise intervention program in an XR environment that is customized for each individual and enables self-training of gait and posture for the promotion of health, and had 17 subjects practice the program. Expert evaluated the gait videos before and after the training and confirmed that all subjects improved their gait. The experts' evaluations provided useful suggestions for information presentation control during the XR experience, and experts' comments provided new insights, such as differences in the improvement effect depending on the physical characteristics of the subjects.

Index terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

In recent years, there have been remarkable advances in VR-related technologies, and there have been various studies on the improvement of gait exercise using VR and XR technologies, but

most of them are aimed at rehabilitation for stroke patients, and there are few examples of gait training to improve health [2].

We have been conducting research on gait and posture improvement in collaboration with judo therapists, who is expert in gait and posture, based on the policy that gait based on correct posture in childhood is important to stay healthy for as long as possible[1]. At present, we know the important evaluation items for improving walking posture [1], and the purpose of this paper is to improve walking posture for health by sensing gait and developing an exercise program to improve gait based on these evaluation items.

2 EVALUATION ITEMS OF GAIT

The gait elements to be improved in the exercise program to be developed are “uni or biaxial,” “arm swing height,” “arm swing width,” and “the whole”. Expert mentioned that “uni or biaxial” is the most important element in the evaluation of foot movement. The “arm swing height” is a measure of the highest point of the fist during gait. “Arm swing width” is a measure of the width of the arm swing trajectory when the gait is viewed from the front. “the whole” section evaluates the overall posture of the gait, including whether the other three evaluation items are well balanced.

3 THE DESIGN OF XR CONTENTS AND TRAINING PROGRAM

We developed contents in the XR space using Unity to improve each of the four evaluation items shown in 2. Figure 1 shows the proposed XR content for interventional gait improvement. Table 1 summarizes the content design for each evaluation item and the exercise intervention program that the subject performs. Table 1 (1) shows the uni or biaxial training, in which two blue guidelines are generated at the user's feet, and the distance between the two lines can be adjusted by the user using the controller to the appropriate biaxial walking foot width as determined by an expert, as shown in the content design diagram, to enable personalized and appropriate biaxial gait training. This enables personalized biaxial gait training.

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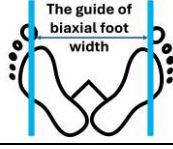
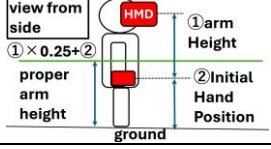
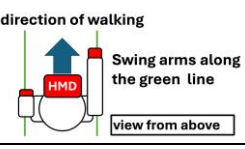
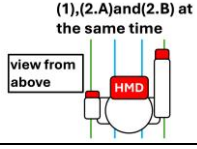
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Table 1. Evaluation items and content design, and the name of the training program

Evaluation items	(1) uni or biaxial footwork	(2.A) Arm swing height	(2.B) Arm swing width	(3) the whole
Contents design				
Training program	Footwork-training	Arm swing-training		Whole-training

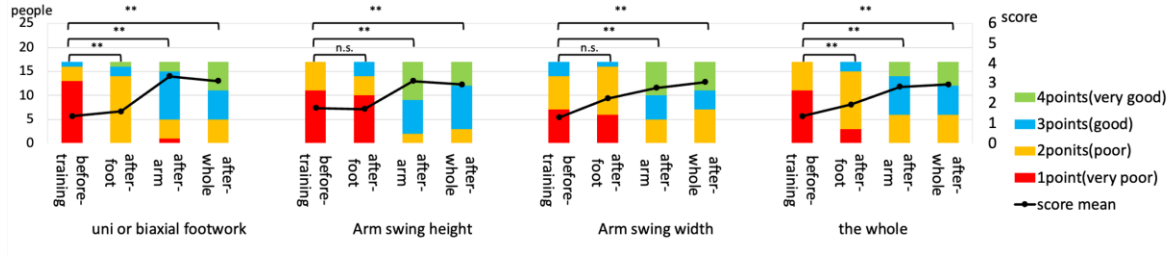


Figure 2: Stacked bar graphs of the results of the four expert ratings and the average change in score for each walk

Walking along the two guidelines displayed at the feet is defined as “footwork-training”. (2.A) is about arm swing height. Previous research in the project has shown that the appropriate arm swing height is the height of the green line illustrated in the content design [1]. This height is calculated based on each person's gait record and is an appropriate guideline for each individual. (2.B) shows the width of the arm swing. The guideline shown in (2.A) is displayed in the direction of travel with two guidelines on the left and right sides, matching the width of each hand. By swinging the arms along the two guidelines, the height and width of the arm swing can be trained simultaneously, and (2.A) and (2.B) are combined as “arm swing-training”. (3) is for the overall evaluation. By displaying all the guidelines (1), (2.A), and (2.B), the user can train arm swing and biaxial footwork at the same time, and this is called the “whole-training”.

4 EXPERIMENT

We conducted an experiment with students at Shizuoka University to verify the usefulness of the content. Seventeen students participated in the experiment ($N=17$, age: 18-23, $M=21.8$, 10 males, 7 females). In the experiment, after the initial gait recording, the three training exercises (footwork, arm swing, and whole training) presented in section 2 were performed sequentially, and the gait was recorded after each training session. And we asked expert to evaluate each video by scoring the four items on a scale of 4 points for “very good”, 3 points for “good”, 2 points for “poor”, and 1 point for “very poor”. In addition, we asked him to comment freely on each walking video from his point of view.

5 RESULT AND DISCUSSION

Figure 2 shows a stacked bar graph of the percentage of four-point ratings by the experts. For each bar graph, the average of the experts' evaluation scores was calculated, and a line graph showing the amount of change in scores for each of the three training sessions is also included in the figure. For the five subjects who had little effect on gait improvement, all four evaluation items were found to have decreased in all subjects' gait records after the overall training. All five subjects who showed poor gait improvement in response to the program showed a decrease in all four evaluation items in their gait recordings after the overall training session. And by matching the scores with the experts' comments, it was found

that subjects with X-legs showed more improvement in content than subjects with O-legs. In Figure 2, the average scores of the experts in the target evaluation items all improved after each training session. This suggests that the exercise intervention content in the XR space developed in this study improved the scores of the experts' evaluation and the subject's gait. In addition, the fact that all the ratings decreased in the gait recordings after the overall training suggests that practicing gait with both hand and foot guidelines in mind distracted the subject's attention, resulting in a decrease in the ratings of the four evaluation items. This result suggests that the content itself needs to be improved after further detailed examination.

6 CONCLUSION

In this study, we proposed a gait improvement exercise program in four evaluation items of gait and posture and developed it as contents in XR space. As a result of the exercise practice using the developed content, the expert evaluation scores for all subjects' gait were improved when compared before the intervention and after the whole training, confirming the improvement of gait and posture. A detailed analysis of the subjects who showed little improvement in the exercise program developed in this study revealed that all the subjects' evaluations decreased before and after the “whole-training” suggesting that training different parts of the body simultaneously in the XR space may cause confusion of consciousness and have a negative effect on the overall gait. This suggests that the control of the information shown to the subjects should be studied and that there is room for improvement of the content itself. For future works, we will expand the target generation and verify the effectiveness of the contents for a wide range of age groups.

REFERENCES

- [1] Akinori Kunishima, Koki Suzuki, Atsushi Omata, Shogo Ishikawa, Shinya Kiriyama, Gait Condition Assessment Methods for Visualizing Interventional Expertise by Means of Posture Detection, Human Activity and Behavior Analysis Advances in Computer Vision and Sensors: Volume 2, Md Atiqur Rahman Ahad, Sozo Inoue, Guillaume Lopez, Tahera Hossain, Boca Raton, 2024.
- [2] Ki Hun Cho, Min Kyu Kim, Hwang-Jae Lee, Wan Hee Lee, Virtual Reality Training with Cognitive Load Improves Walking Function in Chronic Stroke Patients, Tohoku J Exp Med. Vol. 236 No. 4 pp. 273-280, 2015.